LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basics of EMI and EMC
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basics of EMI and EMC. Types of testing.
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of EMI/EMC definitions and testing types from IB/PDF; Instructor supervision required for Q&A; verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a communication system?
(b) Define the term 'noise' in a signal context.
(c) Why is signal integrity critical for aircrew operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Electromagnetic Interference • Sources of EMI in aircraft systems • Impact of EMI on avionics performance • Basic principles of Electromagnetic Compatibility • Distinction between emission and susceptibility	L & Ds	PPT, IB	15 min
(b)	• Types of EMI testing procedures • Radiated emission testing methods • Conducted emission testing methods • Susceptibility testing protocols • Standard compliance requirements for aircraft	L & Ds	PPT, PDF	20 min
(c)	• Shielding techniques for cable bundles • Grounding and bonding best practices • Filtering components in power systems • Common mode chokes application • Maintenance checks for EMC integrity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between EMI and EMC in aircraft systems.
(b) Identify two common sources of interference in a cockpit environment.
(c) Describe the purpose of shielding in reducing electromagnetic coupling.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of EMC discussed in this lesson apply to the upcoming Sonar system introduction in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to sonar system. Purpose and technical details of active and passive sonar system.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min